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Monsoon, 2025

B. Tech 3rd Semester Examination

Concrete Technology

Course Code: BCE23304T

Full Marks –50

Time –2hours

The figure in the margin indicates full marks for the questions.

Section – 1

Q1. Multiple Choice Questions (All compulsory)

10x1=10

1. The specific gravity of coarse aggregate generally ranges between:
a) 2.1–2.3 b) 2.6–2.7 c) 2.5–3.0 d) 3.1–3.5
2. As per IS 269:2015, OPC 43 grade cement has a minimum compressive strength of ____ MPa after 28 days.
a) 33 b) 43 c) 53 d) 63
3. The main ingredient responsible for the initial setting of cement is:
a) C₂S b) C₃S c) C₃A d) C₄AF
4. The workability of concrete primarily depends on:
a) Water-cement ratio b) Curing period c) Aggregate shape d) Mixing time
5. Light-weight concrete is generally used for:
a) Heavy foundations b) Thermal insulation c) Pavements d) Marine works
6. The most common admixture used to increase workability without increasing water is:
a) Retarder b) Accelerator c) Plasticizer d) Air-entraining agent
7. GGBS is obtained as a by-product from:
a) Copper industry b) Steel industry c) Cement industry d) Fertilizer industry
8. In the ACI method of mix design, the water-cement ratio is based primarily on:
a) Durability b) Compressive strength c) Workability d) Cost
9. The rebound hammer test gives an estimate of:
a) Density b) Tensile strength c) Compressive strength d) Modulus of elasticity
10. The concrete used in underwater construction is known as:
a) High density concrete b) Vacuum concrete c) Tremie concrete d) Pumped concrete

Section – 2

Q2 (a). Explain the properties of coarse and fine aggregates and discuss their influence on the strength and durability of concrete. 8

OR

Q2 (b). Describe different grades of cement with their uses and mention relevant IS codes for each type. 8

Section – 3

Q3 (a). Discuss the use of industrial by-products like Fly Ash, GGBS, and Silica Fume as supplementary cementitious materials. 8

OR

Q3 (b). Explain how sustainability can be achieved in concrete construction using novel materials and techniques. 8

Section – 4

Q4 (a). Design an M30 grade concrete mix using IS 10262:2019 method and specify the steps involved. 8

OR

Q4 (b). Explain the concept of Self-Compacting Concrete and describe its advantages, properties, and mix composition. 8

Section – 5

Q5 (a). Differentiate destructive and non-destructive tests. Explain the working principle of the Rebound Hammer and Ultrasonic Pulse Velocity tests. 8

OR

Q5 (b). Describe various NDT methods as per IS 13311 and explain their significance in assessing existing concrete structures. 8

Section – 6

Q6 (a). Explain the maximum and minimum requirements of preparation of concrete under several exposure conditions. Define any three exposure conditions. 8

OR

Q6 (b). Describe different types of chemical admixtures used in concrete and their effects on properties of fresh and hardened concrete. 8
